

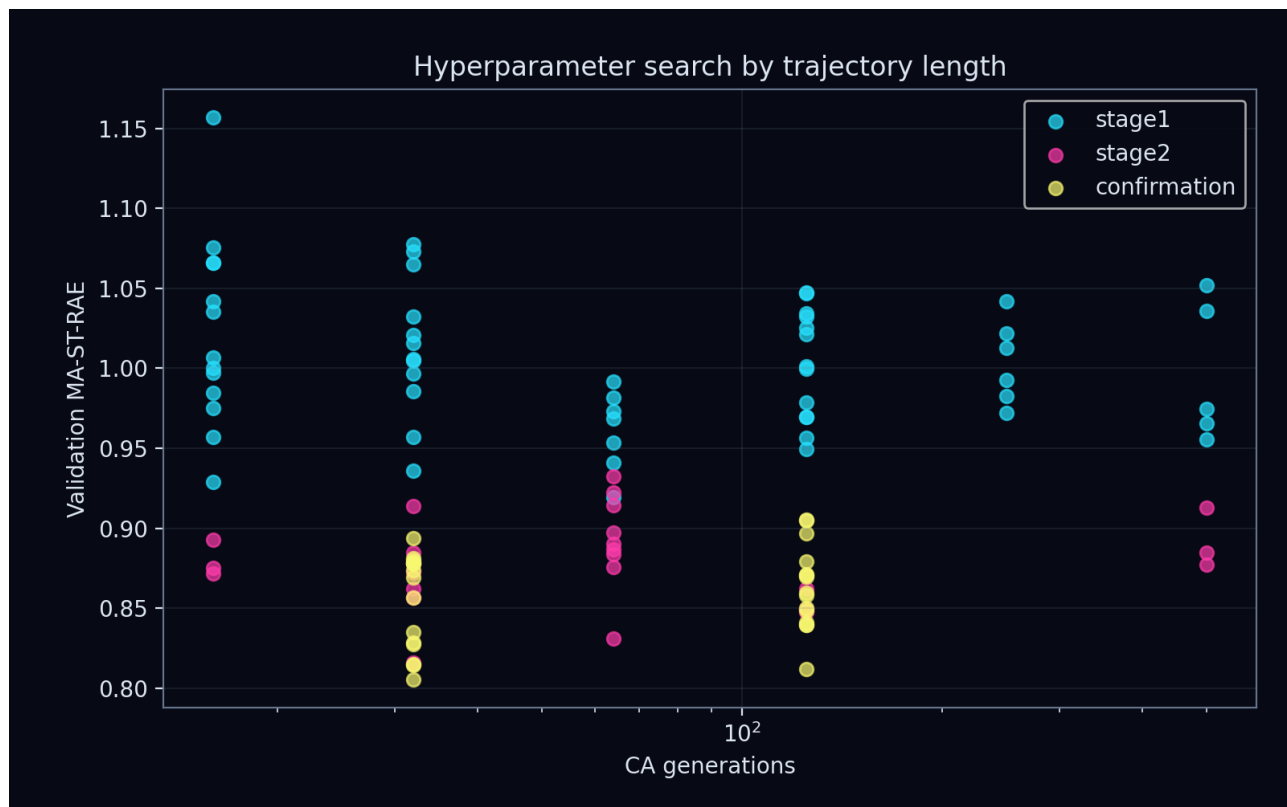
# Strange Matter Engine

## Production study: inertial\_reaction\_diffusion

Direct-inhibition pIC50 | grouped validation | blinded set excluded

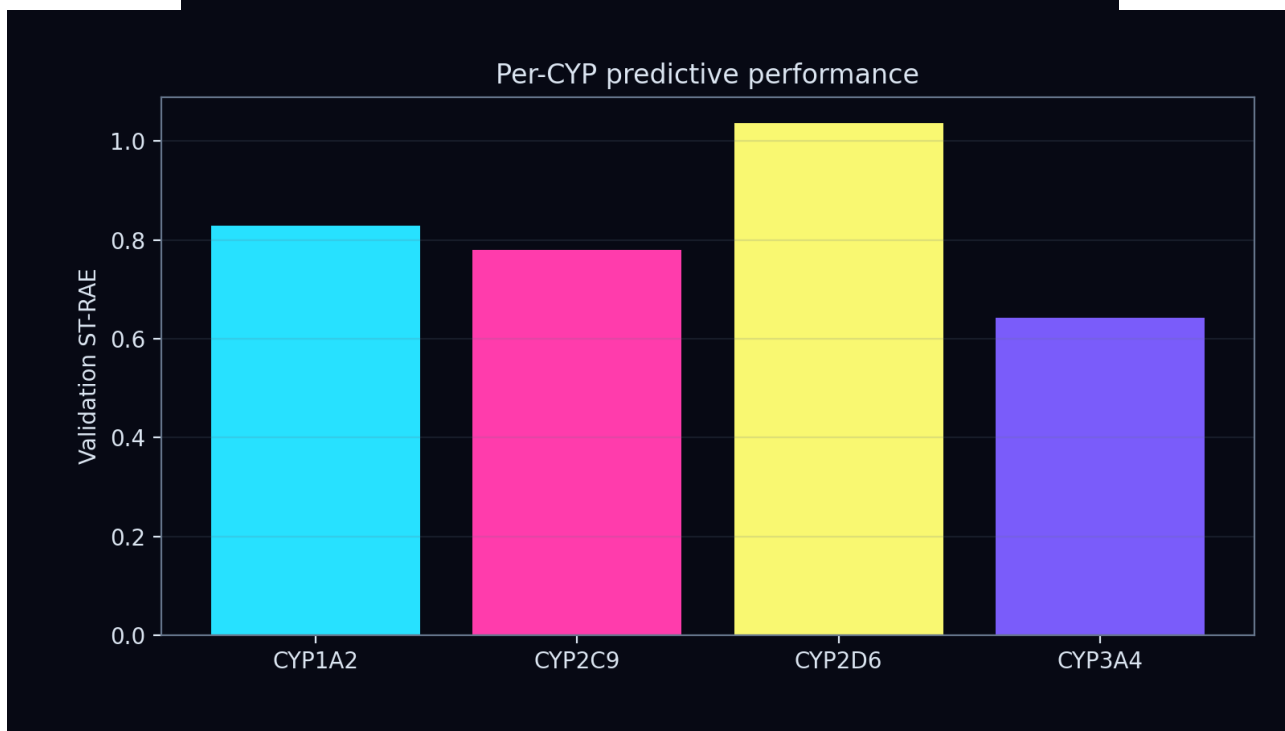
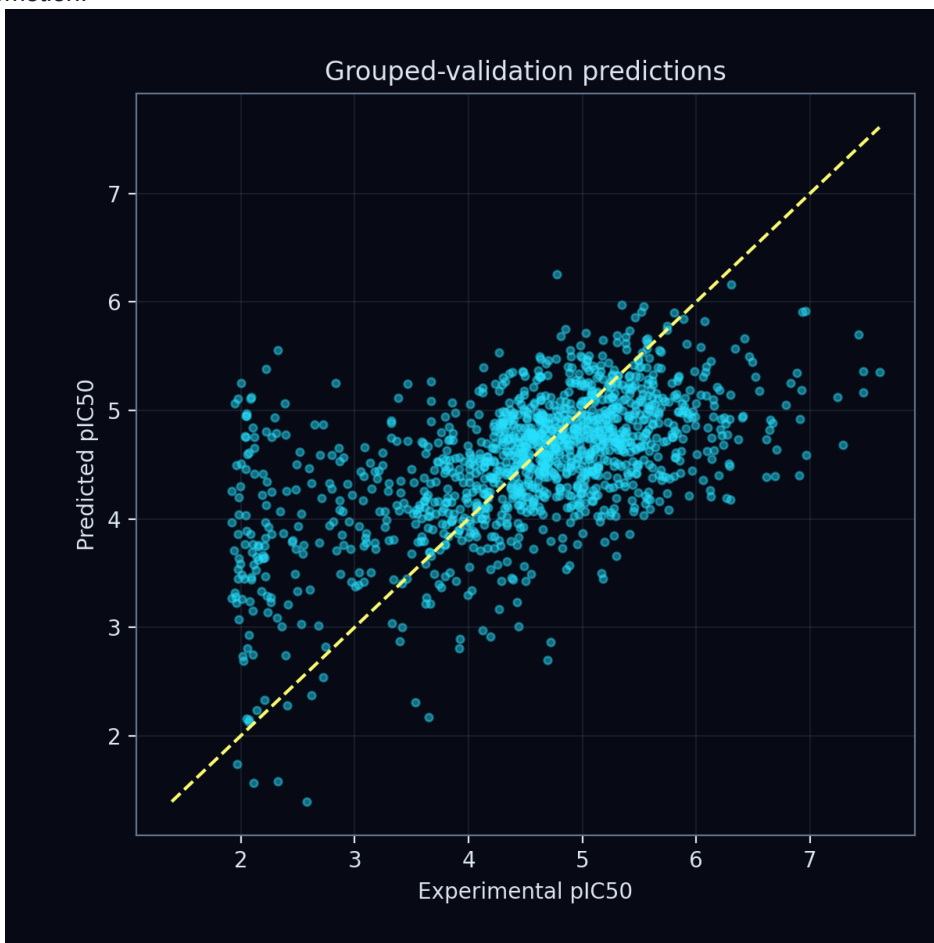
| Quantity                  | Result                  |
|---------------------------|-------------------------|
| Validation MA-ST-RAE      | 0.8252                  |
| Fit MA-ST-RAE             | 0.8298                  |
| Validation RMSE           | 0.8895 pIC50            |
| Fit RMSE                  | 0.8288 pIC50            |
| Generations               | 32                      |
| Dynamical channels        | 16                      |
| Atom feature profile      | electronic              |
| CA learning rate          | 0.003                   |
| Ridge penalty             | 1.0                     |
| CA L2                     | 0.0001                  |
| Gradient clip             | 1.0                     |
| Confirmation mean         | 0.8542                  |
| Confirmation seed SD      | 0.0289                  |
| Bootstrap MA-MAE          | 0.6581 [0.6227, 0.6912] |
| Bootstrap MA-R2           | 0.2291 [0.1845, 0.2753] |
| Bootstrap MA-Spearman rho | 0.4693 [0.4229, 0.5150] |
| Bootstrap MA-Kendall tau  | 0.3326 [0.2990, 0.3674] |

Enhanced CA controls: update scale 0.08, initial-state scale 0.5, support fraction 0.6, bond temperature 1.0. Rule dynamics: A=0.8, B=0.8, C=0.8, D=0.15.



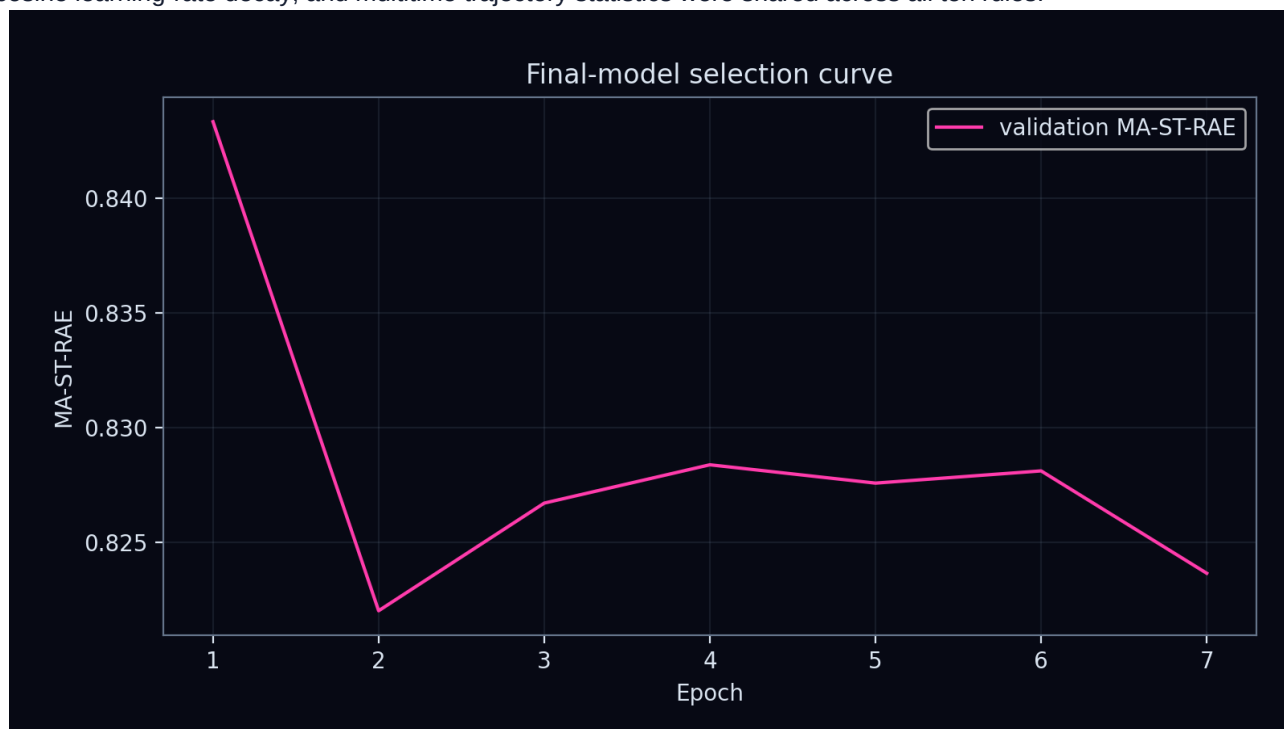
## Predictive validation

Model selection used grouped-validation MA-ST-RAE, the challenge primary metric. Dynamical-interest scores did not influence promotion.



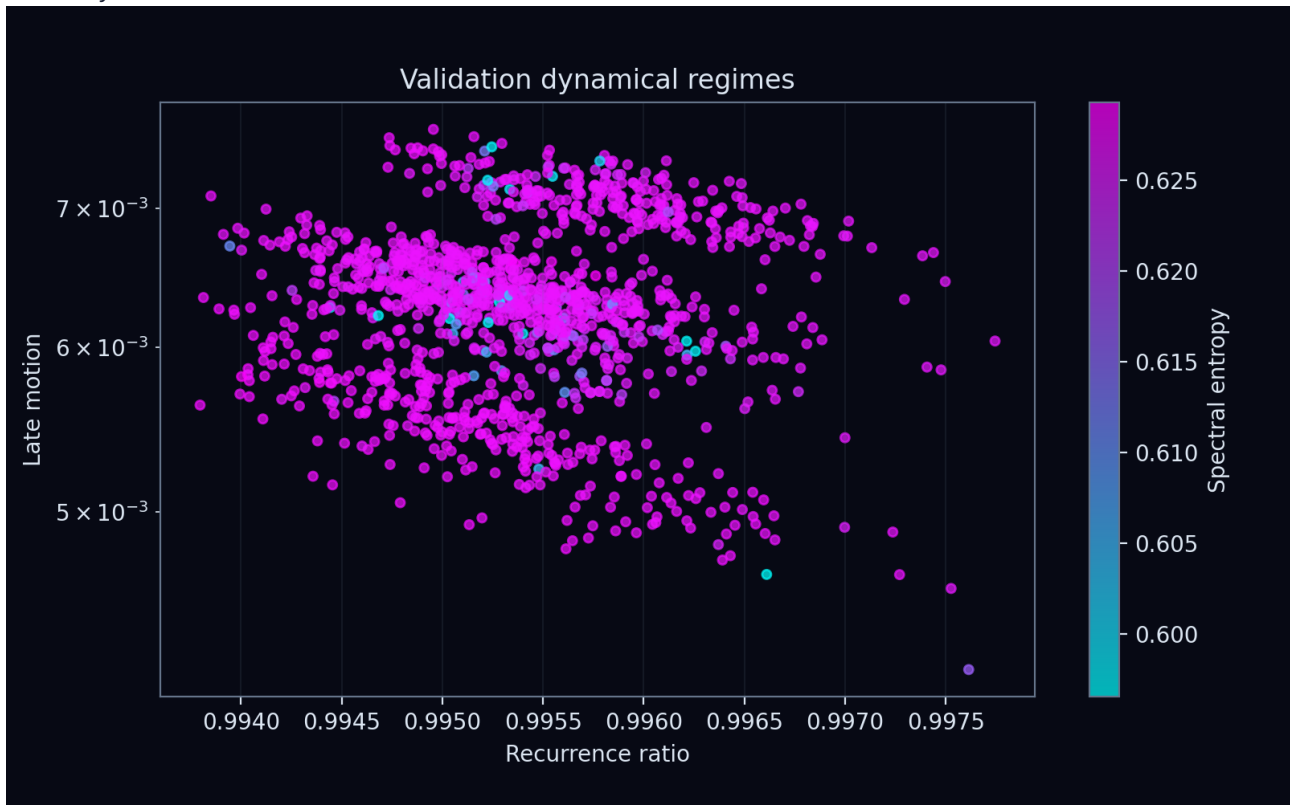
## Optimization

Ridge coefficients and the unpenalized intercept were solved from permitted fitting fingerprints. Query error differentiated through the solve into the graph CA; Adam updated only CA parameters. Bond-conditioned messages, cosine learning-rate decay, and multitime trajectory statistics were shared across all ten rules.



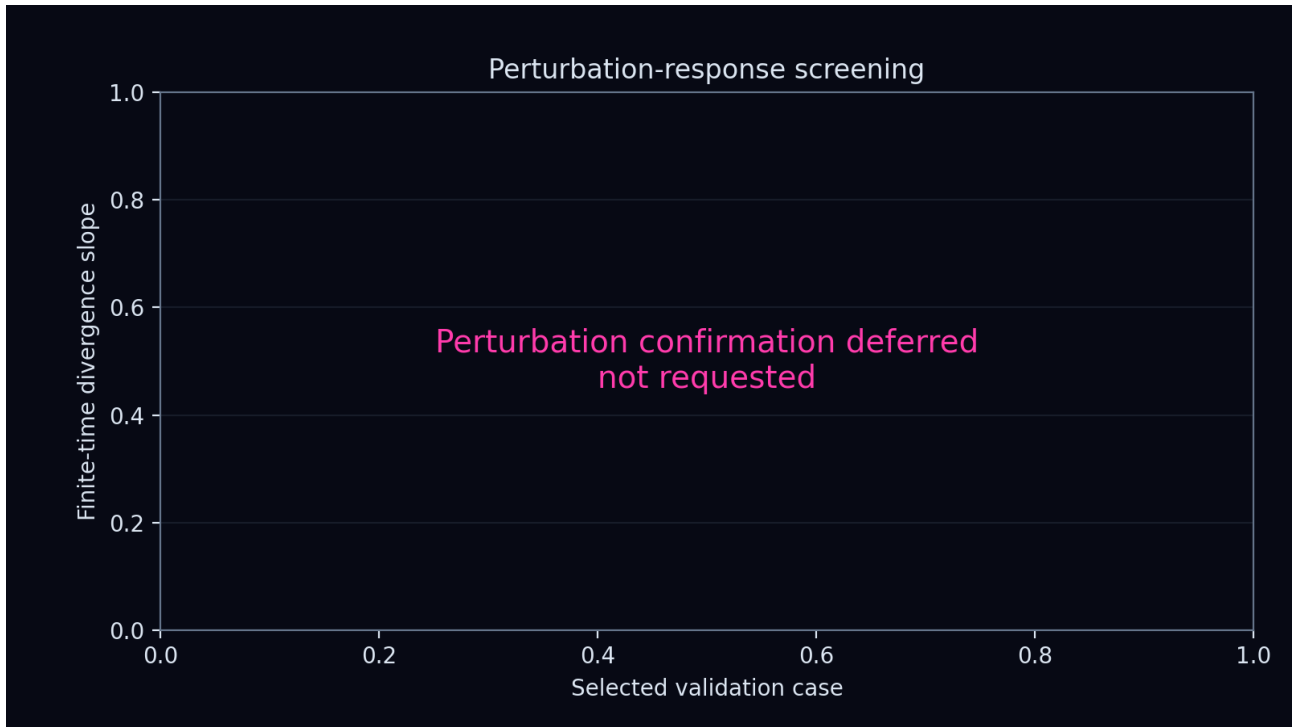
## Emergent dynamics

The validation trajectories were screened for convergence, recurrence, persistent motion, spectral complexity, and perturbation sensitivity. Labels ending in candidate require longer, renormalized and seed-repeated confirmation before any attractor or chaos claim.



## Permutation and perturbation screening

This figure tests whether small state perturbations separate or contract over the observed finite trajectory. It is placed on a dedicated page to preserve its axes, labels, and complete plotting area.



## Scientific limitations

This report describes one transition-rule study under the declared grouped split. Hyperparameter selection and early stopping used labelled training data only. The blinded challenge set was not loaded or predicted. Finite-time positive slopes are screening signals rather than proof of a strange attractor or sustained chaos.